

Configuration-of-Configuration as Substrate Content

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This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

Paper 3 Claim 5 establishes that the configurable dimensions of a Full Aspect Integration (FAI) event are substrate content — authored, inspectable, modifiable, and overridable under the three rights. This note formalizes a structural property of that commitment that goes one level deeper: not only are the FAI configuration dimensions themselves substrate content, but the governance framework specifying *how those dimensions are authored and approved* is also substrate content. Configuration governs FAI events; the governance of that configuration is itself governed. This recursive property — configuration-of-configuration as substrate content — is not a fresh philosophical commitment but an architectural consequence of applying the substrate-content commitment to the configuration layer. The recursion is finite: it bottoms at human-authored governance authority, the same stopping condition Paper 1 Claim 3 establishes for governance generally. The note states the recursive property precisely, describes the three-level structure it produces, explains why the recursion is architecturally necessary rather than merely desirable, traces its inheritance from Paper 2 Claim 6’s recursive applicability commitment, identifies four failure modes it guards against, and provides an operational test for its presence in a deployed system.

1. The sub-commitment stated

D1.23 formalizes the following property of Paper 3 Claim 5:

The governance framework specifying how FAI configuration dimensions are authored and approved is itself authored substrate content, subject to the three rights — inspect, modify, override — at all times, with the recursion bottoming at human-authored governance authority.

This sub-commitment is distinct from D1.22 (the six FAI configuration dimensions as substrate content). D1.22 establishes that the objects being configured — sharing scope, cardinality, persistence policy, cooperation/competition variant, provenance carry-over depth, and multi-mediator coordination — are substrate content. D1.23 establishes that the *process and authority structure* that determines how those objects are written, whose approval is required before they take effect, how they may be changed between FAI events, and what defaults apply in their absence — that entire governance framework is also substrate content.

The distinction is between *what is configured* and *how configuration is governed*. Both are substrate content. Neither escapes the substrate governance boundary.

2. What configuration-of-configuration means in practice

To make the sub-commitment concrete, it helps to distinguish what each level of the recursive structure actually specifies.

The FAI dimensions (Level 1) specify what a given FAI event will do: which aspects are in scope for exchange, how many participants are involved, what happens to the exchanged content after the event concludes, whether the coordination pattern is cooperative or competitive, how deep the provenance record runs, and how multiple substrate mediators are coordinated during the event. These specifications are substrate content — the claim D1.22 establishes.

Configuration governance (Level 2) specifies how the Level 1 dimensions are authored and approved: who holds authority to write each dimension's specification for a given FAI event, what approval process must be completed before the configuration takes effect, how changes to a dimension's specification between consecutive FAI events are authorized, and what default values govern any dimension left unspecified. This governance framework is itself substrate content — the claim D1.23 establishes.

Meta-governance (Level 3) specifies how the Level 2 governance framework is itself authored and approved: who has authority to revise the approval process, how changes to the configuration governance framework are authorized, and what defaults govern the governance framework itself when no explicit specification exists. This level is also substrate content, by the same argument that makes Level 2 substrate content — applying the substrate-content commitment to the governance-of-governance layer produces the same

result.

Human-authored authority (the bottom) is where the recursion stops. At some level, a human governance practitioner authored the foundational configuration framework. That act of authorship is not itself subject to a higher substrate-content commitment requiring its own substrate-content governance layer. The recursion grounds in human governance action, not in an indefinitely extending chain of meta-layers.

The stopping condition is not arbitrary. It is the same commitment Paper 1 Claim 3 establishes for governance generally: governance is an authority architecture, and authority originates in human actors. Applying this commitment recursively through the configuration stack produces the same stopping condition at every level — governance bottoms at human authority, not at protocol invariants, infrastructure defaults, or institutional procedures that operate outside the substrate.

3. Why the recursion is architecturally necessary

The recursive governance property is not a design preference. It is the commitment that forecloses a specific structural vulnerability: *configuration escape*.

Without recursive governance, the following situation is architecturally possible. The six FAI dimensions are substrate content — inspectable, modifiable, overridable under the three rights. But the process that determines how those dimensions are written could itself operate outside the substrate governance boundary. The approval workflow for configuration changes could live in an email thread. The authority structure governing who can write each dimension could be a verbal agreement. The defaults applied when a dimension is unspecified could be a platform setting outside the substrate. In this situation, the configuration objects themselves are governed; the governance of the configuration is not.

The consequence is that the governed substrate contains objects whose origins are not fully traceable within the substrate governance structure. An observer inspecting the substrate can find the configuration specifications, but cannot find, within the substrate, the governance record that authorized those specifications to exist. The configuration looks governed — the objects are present as substrate content — but the authority that produced them is opaque.

Configuration escape is not a theoretical concern. It is the natural resting state of systems that treat governance as a review workflow applied to outputs, rather than as an authority architecture that applies to every layer from which outputs originate. A system that adds substrate-content discipline at the object level without extending that discipline to the configuration-governance level has applied governance partway and stopped.

The recursive governance commitment prevents this by extending the substrate-content commitment upward through the configuration stack without exception. Every layer from which configuration originates — including the governance framework that authorizes configuration, and the meta-governance framework that authorizes the governance framework — is substrate content. There is no layer at which the substrate-content commitment terminates before reaching human-authored authority.

This is why the recursion is an architectural commitment rather than a design preference. A design preference can be overridden when convenient. An architectural commitment holds as a structural invariant because violating it creates the specific vulnerability the commitment was introduced to foreclose.

4. Inheritance from Paper 1 Claim 3 and Paper 2 Claim 6

The recursive governance property does not originate at Paper 3. It is the consequence of applying two prior commitments at FAI configuration scope.

From Paper 1 Claim 3 (human-governed authority): All substrate content — including the orchestration rules that govern how other substrate content is authored — is subject to the three rights at all times. This commitment applies directly to configuration governance: the framework specifying how FAI dimensions are authored is an orchestration rule governing the authoring of substrate content. Under Paper 1 Claim 3, that framework is itself substrate content, subject to the three rights. Paper 1 Claim 3 does not require configuration specifically to be present — it applies to orchestration rules as a class, and configuration governance rules are a member of that class.

The authority-vs.-labor distinction Paper 1 §3.3 makes portable is equally active here. Configuration governance is an authority architecture, not a review workflow. Humans hold authority over the governance framework; substrate mediators — including AI systems operating within the substrate — draft, propose, and populate configuration governance specifications, but do not hold authority over them. The authority-vs.-labor distinction applies at the configuration-governance level in the same way it applies at the substrate-content level.

From Paper 2 Claim 6 (recursive applicability): Paper 2 explicitly established that governance over governance is also substrate content — that the commitments Paper 1 introduces apply not only to the substrate objects those commitments govern, but also to the governance structures that determine how those objects are authored and modified. Paper 2 Claim 6's recursive applicability commitment is the structural antecedent for D1.23: the recursion Paper 2 establishes within a single Self applies at the inter-Self scope when Paper 3's shared substrate is present.

The inheritance edge is: P2-C6 → P3-C5 → D1.23. Paper 2 Claim 6 establishes recursive applicability within a Self's substrate governance. Paper 3 Claim 5 inherits that structure and extends it to FAI configuration scope — the shared substrate during FAI events, operating under joint governance authority across participating Selves. D1.23 formalizes the recursive property as a named sub-commitment of Paper 3 Claim 5, tracing the inheritance chain back to Paper 2 Claim 6 and, through it, to Paper 1 Claim 3.

What Paper 3 adds to the inherited structure is the joint-authority dimension. At the inter-Self scope, the governance framework for FAI configuration is not authored under a single participating Self's authority but under the joint authority of all participating Selves. Configuration-of-configuration as substrate content, at the FAI scope, means that the joint-authority governance framework is itself jointly authored substrate content — all participating Selves retain the three rights with respect to the governance framework, not only with respect to the configurations that framework produces.

5. Four failure modes the sub-commitment defends against

Failure mode 1 — Configuration process outside the substrate. The configuration governance framework for FAI events operates through mechanisms that are not substrate content: email approval threads, verbal agreements among governance participants, platform-native workflow tools whose state is not exposed as inspectable substrate content. In this failure mode, the six FAI configuration dimensions are substrate content but the process that authorized them is not. An observer can find the configuration objects but cannot trace their authorization within the substrate governance structure. The recursive governance commitment prevents this by requiring the approval and authorization machinery to be substrate content, not only the outputs that machinery produces.

Failure mode 2 — Non-inspectable meta-governance. The governance framework for FAI configuration is itself present as substrate content, but the governance framework that governs *that* framework — how the configuration governance framework may be changed, who holds authority to revise it, what defaults apply when it is unspecified — is not inspectable under the three rights. A participating Self or governance practitioner can read the Level 2 governance framework but cannot read, modify, or override the Level 3 meta-governance framework that determines how Level 2 may evolve. The recursive governance commitment prevents this by extending the three rights through every level of the governance stack, not only through the first level above the configuration objects.

Failure mode 3 — Infinite meta-governance regress. The recursion continues indefinitely without bottoming at human-authored authority. Each meta-governance level requires a further meta-meta-governance level, producing an architecture with no stable foundation. This

failure mode typically arises when the bottom-out condition is misidentified — placed at a protocol invariant, an institutional procedure, or a technical default rather than at human governance action. The recursive governance commitment prevents this by specifying the stopping condition precisely: the recursion bottoms at the level at which a human governance practitioner authored the foundational governance framework. Below that level, there is no further substrate-content governance layer to apply — there is only human authority, which is the origin of governance rather than an object of it.

Failure mode 4 — Flat configuration. The FAI configuration dimensions are treated as substrate content, but configuration-of-configuration is treated as infrastructure default — a platform convention, a vendor setting, or a protocol invariant outside the substrate governance boundary. This failure mode presents as compliance at Level 1 with non-compliance at Level 2: the configuration objects look governed because they are present as substrate content, but the governance framework that produced them is not itself subject to the three rights. An observer examining the substrate cannot find the configuration governance framework as authored, inspectable substrate content; they can only find the configurations it produced. The recursive governance commitment prevents this by treating the distinction between “configuration as substrate content” and “configuration-of-configuration as infrastructure default” as an architectural error, not a permissible design choice.

6. Operational test

A system instantiates the D1.23 configuration-of-configuration commitment if and only if all of the following conditions hold:

Test condition 1. For a given FAI event's configuration, an observer with appropriate access can locate within the substrate — as authored, persistent substrate content — not only the six configuration dimension specifications (D1.22) but also the governance framework specifying how those dimensions were authored and approved: the authority structure, the approval process, and any applicable defaults.

Test condition 2. The governance framework located in Test condition 1 carries provenance within the substrate: it is possible to determine who authored it, when it was authored, what approval it received before taking effect, and whether it has been modified since its initial authoring.

Test condition 3. The three rights apply to the governance framework as substrate content: a human with appropriate access can inspect the governance framework without scheduling or intermediation, can modify it with the change taking effect as substrate state, and can override any aspect of it. No architectural layer — platform, vendor, runtime middleware, or AI

system — can in principle prevent the exercise of these rights.

Test condition 4. The same three conditions hold at the next level up: the meta-governance framework specifying how the Level 2 governance framework may be modified is itself present as inspectable, modifiable, overridable substrate content with provenance.

Test condition 5. The recursion bottoms at a level at which the foundational governance framework was authored by a human governance practitioner, with that authorship itself present in the substrate record as the origin point of the governance stack.

A system that passes Test conditions 1–2 but fails Test condition 3 has configuration governance as substrate content but not under the three rights — a substrate record without governance. A system that passes Test conditions 1–3 but fails Test condition 4 has one-level recursion rather than full recursive governance — compliance at Level 2 with flat configuration at Level 3. A system that passes Test conditions 1–4 but cannot demonstrate Test condition 5 has a recursive governance structure without a grounded stopping condition, leaving the governance stack’s authority origin untraced.

The full commitment requires all five conditions.

7. Conclusion

Configuration-of-configuration as substrate content is the recursive consequence of applying the substrate-content commitment to the governance layer — the layer that governs how FAI configuration dimensions are authored and approved. The recursion is not a philosophical extension of the architecture; it is the structural result of applying the commitments Paper 1 Claim 3 and Paper 2 Claim 6 establish to the configuration governance context Paper 3 introduces.

The stopping condition — human-authored governance authority — is the same stopping condition that grounds governance generally in the CKS trilogy. Governance is an authority architecture whose origin is human action; the recursive application of the substrate-content commitment terminates at that origin rather than continuing into infinite regress.

The architectural value of the commitment is the foreclosure of configuration escape: the structural vulnerability in which configuration objects are governed but the governance of those objects is not. By requiring the governance framework itself to be substrate content, subject to the three rights with provenance, the recursive governance commitment ensures that no layer of the configuration stack escapes the substrate governance boundary. Every layer is inspectable. Every layer is modifiable. Every layer is overridable. The boundary holds all the way down to human-authored authority.

Downstream work formalizing operational variants of recursive governance, or anti-patterns in which the recursion is shallow rather than fully grounded, should take D1.23's stopping condition — human-authored authority — as the reference point for evaluating whether a proposed recursion depth satisfies the architectural commitment.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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